Firmware Supplement

Package format, ROM contents and undocumented commands

Community reverse-engineering supplement for firmware A.03.03
Document 344XXA-90S01
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Scope

This document supplements, and does not replace, the Keysight Truevolt Series Operating and Service Guide (Keysight 9018-03876), which covers all four models. It describes how the 3446xA firmware update package is built, what the instrument's ROM contains, and the commands and settings found in the firmware that the Operating and Service Guide does not list. It does not cover, and gives no procedure for, licensing, options or instrument security.

Provenance

Everything here comes from static analysis of the firmware update file Agt3446XA_rev_A_03_03.upd. The package was unpacked, the ROM image was decompressed, and the resulting modules and files were examined. Command names come from the firmware itself and from Keysight programs in the ROM that send them. What a command does is inferred from the surrounding names, messages and menus unless the text says otherwise. No command in Chapter 5 was sent to an instrument for this revision.

Applicability

Firmware revision A.03.03, built 2 July 2021, a single package for all four models. Only this revision was examined. Other revisions may use different formats and command sets.

Conventions

SCPI keywords appear in the short form the firmware stores; the long-form letters are understood. Hexadecimal values take a $ prefix. Command text and file names are set in a monospaced face. The factory manual is cited as the Operating and Service Guide.

Disclaimer

This is an independent document. It is not a Keysight product, carries no Keysight endorsement, and is not warranted to be correct or complete. Several commands described here change internal state. Do not send them to an instrument unless you understand the consequence and can restore the previous state. The authors accept no responsibility for damage, lost calibration or a voided warranty.

Acknowledgement and preparation

Prepared by Jared Cabot (jetstreamtechnology@protonmail.com). The firmware analysis, the extraction tools and the text were prepared with Claude (Anthropic). The LZX decoder used by the tools is libmspack by Stuart Caie. This document is released under the MIT license. The firmware it describes is the property of Keysight Technologies and is not reproduced here.

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1

General Information

Instrument Family

The Keysight Truevolt series is a family of bench digital multimeters. The 34460A, 34461A and 34465A are 6½-digit instruments and the 34470A is a 7½-digit instrument (Operating and Service Guide, Introduction to the Instrument). All four run the same firmware: one update package, `Agt3446XA_rev_A_03_03.upd`, carries the firmware for every model, and the running instrument selects its behaviour from its model and installed options.

What the Firmware Package Contains

The package holds a Windows Embedded CE 6.0 operating system image for an ARM processor. Inside it are 186 execute-in-place modules (the operating system, drivers and native support libraries) and 253 files. The multimeter application is `Torreys.dll`, a native ARM program of 8.6 MB stored as a compressed ROM file. Above it sits a .NET Compact Framework 3.5 layer for the command parser framework, the LXI web services and the graphical front panel.

How to Read This Supplement

Chapters 2 and 3 describe the package and its compression, so the image can be rebuilt from the update file. Chapter 4 describes the ROM contents. Chapter 5 lists the commands and settings that the Operating and Service Guide does not document. Chapter 6 gives the toolchain, and Chapter 7 records what was and was not verified, and what changed from Revision 1.0.

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Firmware Package Format

The update file is three formats nested inside one another. All multi-byte values are little-endian.

The .upd Wrapper

The outermost layer is a length, an XML description, and a gzip stream:

```
offset 0        u64   length of the XML text
offset 8              XML description (UTF-8)
offset 8+len          gzip stream, to end of file
```

The XML names the four models, the version (`A.03.03`), the build date, and a base64 MD5. The MD5 covers the inflated gzip contents, which is the XPRESS container described next, not the final ROM image.

The XPRESS Container

The container begins with a four-byte signature and the total output size:

```
'XPRS'  (4 bytes)
u32     total uncompressed size
then repeated blocks:
  u32   block size, with flags in the top two bits
  ...   block data
```

Bit 31 of the size word marks a stored block, copied out verbatim. Bit 30 marks a short final block, where the next word is the output length; the encoder leaves the last four output bytes implicit, so the decoder zero-fills to that length. Any other block is a Microsoft XPRESS LZ77 stream that expands to 65536 bytes. The output is the ROM image `nk.bin`.

The Windows CE ROM Image

The file `nk.bin` is a standard Windows CE record image: the signature `B000FF`, a load address and a length, then records of target address, length, checksum and data. A record with a zero address ends the list and carries the entry point. All 168 record checksums pass, which confirms the XPRESS decode.

Writing each record to its address gives a flat image based at `$80361000`. The `ECEC` signature at offset `$40` points to the ROM header, which lists the 186 modules and 253 files. The CPU type in the header is `$01C2`, the PE machine type Windows CE uses for ARMV4I code: 32-bit ARM instructions with Thumb interworking. The kernel identifies itself as built on 8 March 2013.

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ROM File Compression

Modules keep their executable code uncompressed so it can run in place. Their writable data sections, and most ROM files, are compressed: the `$800` file attribute marks a compressed file, and `$2000` a compressed module section. This compression was the main obstacle to reading the firmware, so it is given in full.

Page Layout

A compressed item is divided into 4096-byte pages. It begins with a table: a three-byte total uncompressed size, then one three-byte entry per page giving the offset, from the start of the item, at which that page's block ends. Page zero's block begins immediately after the table, and each later block begins where the previous one ended.

The Per-Page Block

Each block starts with a 16-byte header, followed by the compressed data:

```
u32  $00000010         header length, including this word
u32  uncompressed size  (4096, except the final page)
u32  compressed size    (bytes after the header)
u32  uncompressed size  (repeated)
u32  $8D80805B          first word of the compressed data
...  LZX bitstream
```

The word `$8D80805B` is the same on every one of the 5,933 pages in the ROM's files. The decoder steps over it to reach the bitstream.

The Codec

The bitstream is Microsoft LZX with a 64 KB window (a window-bits value of 16), one independent stream per page. After decoding, the Intel E8 call translation has to be undone, with a translation size of 12,000,000 and positions counted from the start of each page. The stream's own E8 header bit is zero on every page, so a standard LZX decoder will not do this by itself.

Caution

Without the E8 step the output looks plausible but is wrong after many `$E8` bytes. In this ROM that affects 73 of the 222 compressed files and 34 module images, including `Torreys.dll` and the .NET assemblies. Revision 1.0 of this document and its tools omitted the step.

The decode is confirmed by integrity checks built into the files themselves. Every PNG image passes its chunk CRCs, the web applet `DM344XXA.jar` passes a full zip CRC test, and every base relocation in `Torreys.dll` has a valid type. Every compressed item also decodes to exactly the size its header declares.

4

ROM Contents

Modules and Files

The ROM holds 186 modules and 253 files. The modules are the operating system, its drivers (file names carry `SH600` and `P500` platform prefixes, and one is named `GpibOverSPI_P500_CE6.dll`) and Keysight's native support libraries. The files include the multimeter application, the .NET assemblies, the web interface, fonts, front-panel firmware and configuration data. The complete inventory is in `analysis/rom_inventory.txt` in the accompanying repository. Table 4-1 lists the parts most relevant to programming and service.

Table 4-1. Components Most Relevant to Programming and Service

File	Role
`Torreys.dll`	Native measurement application and SCPI engine
`Torreys.exe`	.NET host that starts Torreys.dll
`Agilent.Cdf.Api.UI.Scpi.dll`	SCPI parser framework (.NET)
`Agilent.Cdf.Api.Lxi.dll`	LXI services (.NET)
`Agilent.Cdf.Api.Security.dll`	Secure storage and secure erase (.NET)
`Agilent.Cdf.Api.Licensing.dll`	Option and license handling (.NET)
`LxiInstrumentIO.dll`	Web SCPI console back end (ISAPI)
`WarmupRackTest.exe`	Keysight warm-up test tool
`AgtFrontPanelKeyMap.34460-66502.xml`	Front-panel key map
`AgtFPNxp.2.40.hex`	Front-panel firmware, NXP variant
`AgtFPMsp430.3.15.hex`	Front-panel firmware, MSP430 variant

The Measurement Application

`Torreys.dll` holds the instrument's commands and measurement behaviour: the SCPI keyword table, calibration, self-test, and an internal peek/poke layer used by calibration and self-test. It is a standard PE image (machine `$01C2`, image base `$10000000`) with a valid relocation table, so it loads into common disassemblers once extracted. Its debug path names the `STM320s_CE6.0_ACCL (ARMV4I)` build platform.

The Web Interface

The instrument serves an LXI web interface of 36 ASP pages. The SCPI console, `InstrumentIO.asp`, posts to `/InstrumentIO/Execute` and `/InstrumentIO/DeviceClear`, which `LxiInstrumentIO.dll` serves. `WebControlMethod.asp` is the back end of the remote front panel: the page script sends it key presses and SCPI, and it passes them to `WebControlDll.dll`. Three pages, `DataGet.asp`, `DataSend.asp` and `DataTransfer.asp`, are unused templates from an oscilloscope product. They carry an EXAMPLE watermark, describe waveform features, and do nothing on the multimeter.

The Front Panel

The front panel has its own microcontroller, in one of two hardware variants: an NXP P89LPC932A1 (8051) or a TI MSP430FR2033. The ROM carries firmware and a configuration file for each (`AgtFPNxp.2.40.hex`, `AgtFPMsp430.3.15.hex`). At boot, `AgilentFrontPanelSerial.dll` identifies the variant from a measured voltage. Both variants use a serial link and share one key map, `AgtFrontPanelKeyMap.34460-66502.xml`, which lists 28 keys on an 8 by 4 matrix with each key's row, column, label and Windows virtual-key code.

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Undocumented Commands and Features

None of the commands in this chapter appears in the Operating and Service Guide, which documents no `DIAGnostic` or `SYSTem:FPANel` commands at all. Each entry says where the command was found. Where a Keysight program in the ROM sends the command, its syntax is taken from that program; otherwise syntax and meaning are inferred. None was sent to an instrument for this revision.

AC Mode

VOLT:AC:MODE and CURR:AC:MODE

`[SENSe:]VOLTage:AC:MODE <mode>`

`[SENSe:]CURRent:AC:MODE <mode>`

Both commands appear in the instrument's own list of settings it reports when saving state, next to the documented AC bandwidth, null and range settings. The front-panel menus have a Meas Settings entry, AC Mode, with the choices RMS and Avg.

From firmware analysis alone, this selects between the documented true-RMS AC measurement and an average-responding mode that the Operating and Service Guide does not mention. The parameter spellings accepted over SCPI are not confirmed.

Found in Torreys.dll; not in the Operating and Service Guide.

The DIAGnostic Subsystem

The keyword table in `Torreys.dll` holds `DIAG` with `POKE`, `PEEK`, `OFINIT`, `OFCLR`, `FPGA` and `FP/KEYPress`.

DIAG:POKE

`DIAG:POKE <op>[,<a1>[,<a2>[,<a3>]]]`

A numbered internal operation: an operation code followed by up to three arguments. The Keysight warm-up test tool sends `DIAG:POKE 123,1,0`, `DIAG:POKE 123,1,1` and `DIAG:POKE 37`. One operation reports whether the instrument holds a primary and a secondary AC flatness data record. What the other operation codes do is not known.

Caution

The operations are internal and undocumented, and some may change calibration or hardware state. Do not send operation codes you have not identified.

Syntax from WarmupRackTest.exe, which sends it; not in the Operating and Service Guide.

DIAG:PEEK?

`DIAG:PEEK? <a>, <b>`

The read-side companion to `DIAG:POKE`. The warm-up test tool sends `DIAG:PEEK? 2, 4` and `DIAG:PEEK? <n>, 5`. The meaning of the arguments is not known.

Syntax from WarmupRackTest.exe, which sends it; not in the Operating and Service Guide.

DIAG:FPGA:SWITch:SET

`DIAG:FPGA:SWITch:SET <n>`

Sent by the warm-up test tool with a numeric argument. From the name, it sets a switch configuration in the measurement FPGA. Its effect is not known.

Syntax from WarmupRackTest.exe, which sends it; not in the Operating and Service Guide.

DIAG:FP:KEYPress

`DIAG:FP:KEYPress "<key>"`

`DIAG:FP:KEYPress "QuickHelp <key>"`

Simulates a front-panel key press. The firmware has a front-panel event recorder that writes `.demo` macro files made of `run_scpi`, `wait`, `simulate_key` and `DIAG:FP:KEYP` lines. Key names include `Softkey0` to `Softkey5`, `Hardkey0` to `Hardkey29`, `KnobLeft`, `KnobRight` and `EnterNumber`. The `QuickHelp` form triggers a key's quick-help action, the press-and-hold help screen.

Found in Torreys.dll; not in the Operating and Service Guide.

DIAG:OFINIT and DIAG:OFCLR

Factory identity initialisation. `DIAG:OFINIT` checks for a 10-character serial number that begins with two letters and writes the instrument's identity records to its secure storage. Their parameters and behaviour are deliberately not described here. The identity records determine the instrument's model and feature set, which is outside the scope of this document.

Caution

These commands rewrite the instrument's stored identity. Do not send them to a working instrument.

Found in Torreys.dll; not in the Operating and Service Guide.

Commands Used by the Web Interface

The web interface's browser applet, `DM344XXA.jar`, sends these commands to the instrument. Their purposes are inferred from their names and from where the applet uses them.

`DIAG:MON:STATE`	ON, OFF, or query. Turns a monitor mode on and off.
`DIAG:LOCAL`	Returns control to the front panel.
`DIAG:REMOTE?`	Asks whether the instrument is in remote.
`DIAG:WEB:LOG`	Writes to a web-interface log.
`DIAG:WEB:TIME`	Web-interface timing.
`DIAGnostic:APPlication:MODel?`	Returns the model the application is running as.

The remote front panel's server side, `WebControlDll.dll`, presses keys with `SYST:FPAN:KEYP <id>`, a key interface separate from `DIAG:FP:KEYPress`. The .NET application framework also defines a `DIAGnostic:CDF` branch of 28 nodes whose leaves are named after data types; it appears to exist for testing the command framework.

Secondary Measurement Nodes

The firmware's state list has `:SEC` nodes for nine function paths: `CAP`, `CURR`, `CURR:AC`, `FREQ`, `FRES`, `TEMP`, `VOLT`, `VOLT:AC` and `VOLT:RAT`. These are documented in the Operating and Service Guide, which also documents secondary measurements for period and two-wire resistance. They are listed here only because they are easy to overlook.

Internal Mechanisms

The firmware also contains the following. This supplement records only that they exist and gives no procedure for operating them.

- A feature manager with a developer mode and a production-test mode. Developer mode cannot be enabled while the instrument is secured, and production-test mode requires developer mode.
- A secure-code subsystem covering the secure code, a secure count, and the calibration-secured state.
- A boot-ROM operation reachable through the internal peek/poke layer.

6

Reproducing This Work

The Toolchain

The accompanying repository carries a small toolchain that unpacks the update file and extracts the ROM: three Python stages and one C program, the page decompressor, built on libmspack's LZX decoder. It needs Python 3 and a C compiler. Extraction takes about ten seconds.

Steps

Place the update file in `Firmware/` and run, from the repository root:

```
cd tools &amp;amp;&amp;amp; sh build.sh

mkdir -p ../work

python3 unpack_upd.py  ../Firmware/Agt3446XA_rev_A_03_03.upd ../work/nk.bin

python3 b000ff.py      ../work/nk.bin ../work/nk_flat.bin

python3 extract_all.py ../work/nk_flat.bin ../work/extract
```

The first stage checks the package MD5; the second checks every record checksum; the third reports any page that fails to decode. `extract_scpi.py` and `diff_manual.py` then build the command inventory and compare it with the text of the Operating and Service Guide.

The extracted files belong to Keysight and Microsoft and are not redistributed. The tools regenerate them from an owner's own copy of the update file.

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Verification Record

Confirmed by Analysis

- The package MD5 matches the inflated XPRESS container.
- All 168 ROM image record checksums pass.
- All 222 compressed files and all 309 compressed module sections decode to the sizes their headers declare, with no decoder errors.
- All 15 PNG images pass their chunk CRCs, the applet JAR passes a full zip CRC test, and every base relocation in Torreys.dll has a valid type.
- The firmware command inventory holds 148 command paths after folding long-form spellings. Diffed against the Operating and Service Guide, with its optional and alternative nodes expanded, it leaves the five paths described in Chapter 5 (two AC Mode commands and three DIAG commands). The other Chapter 5 commands were found in the programs that send them.

Read From Firmware, Not Yet Exercised

Every command in Chapter 5 was identified from the firmware and the programs that use it, not tested. The names are firm. Parameters, responses and side effects are not confirmed. Testing them needs a spare instrument and a way to restore it.

Not Yet Recovered

The command inventory comes from readable strings. The full command tree, with every node's parameters and any hidden flag, is built in code inside `Torreys.dll` and the .NET parser framework. Recovering all of it needs further analysis of those modules.

Corrections in This Revision

Revision 1.0 was checked by an independent review, which found the following errors. All are corrected here.

- The ROM decompressor omitted the Intel E8 step, so 73 files and 34 module images were extracted with wrong bytes. The text files used to check it contain no `$E8` bytes and could not reveal the fault.
- The package MD5 was said to cover the ROM image. It covers the XPRESS container.
- The operating system was given as Windows CE 7 and the code as Thumb-2. It is Windows Embedded CE 6.0, and the code is 32-bit ARM (ARMV4I).
- The 34460A and 34461A were given as 5½-digit. They are 6½-digit.

- DIAG:POKE was described as writing AC flatness calibration data, DIAG:FP:KEYP as a help-text command, the front panel as using two microcontrollers together, and the key map as listing USB HID codes. Chapters 4 and 5 now give what the firmware supports.